

Polymyxin B haemoadsorption in endotoxic septic shock (Tigris): a multicentre, open-label, Bayesian, randomised, controlled, phase 3 trial



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Summary

Background Endotoxic septic shock, a subset of septic shock with high endotoxin activity and multiorgan failure, is associated with high risk of death. We sought to identify the effect of endotoxin removal from the blood with polymyxin B haemoadsorption on mortality.

Methods We conducted an open-label, randomised, controlled, phase 3 trial at 19 US hospitals, enrolling adults (aged ≥ 18 years) with septic shock requiring vasopressors, with multiple organ dysfunction, and with endotoxin activity between 0·60 and 0·89 units. Patients were randomly assigned (2:1) to receive two sessions of polymyxin B plus standard of care or standard of care alone (control), with block randomisation stratified by site. Study personnel were masked to treatment allocation. Polymyxin B haemoadsorption was given via haemodialysis at a blood flow rate of 80–120 mL/min for 90–120 min per session, 22 h apart. The primary outcome was 28-day mortality in the intention-to-treat cohort. The safety cohort included all participants exposed to any amount of study treatment. Design and analysis followed a Bayesian framework, using a prior for the treatment effect based on a subgroup of the earlier EUPHRATES trial. 90-day mortality was the key secondary outcome. The trial was registered at ClinicalTrials.gov and is completed (NCT03901807).

Findings Between Sept 30, 2019, and April 10, 2025, we screened 14 890 patients, of whom 157 were enrolled (106 assigned to polymyxin B and 51 to control; 66 [42%] women and 91 [58%] men). At 28 days, 41 (39%) patients in the polymyxin B group and 23 (45%) in the control group had died, yielding a posterior probability of benefit of 95·3% (APACHE-II adjusted odds ratio 0·67 [95% credible interval 0·39–1·08]). At 90 days, the posterior probability of benefit was 99·4% (0·54 [0·32–0·87]). Of 100 patients in the polymyxin B group assessed for safety, 30 (30%) had a serious adverse event compared with 11 (22%) of 51 patients in the control group (difference –8 percentage points [95% CI –22 to 6]). Two (2%) serious adverse events in the polymyxin B group were treatment-related, one related to polymyxin B and one to catheter placement.

Interpretation In patients with endotoxic septic shock defined by high endotoxin activity and multiorgan failure, polymyxin B haemoadsorption was associated with a high probability of lower mortality at 28 days and 90 days.

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Introduction

Septic shock is a severe form of dysregulated host response to infection and has a mortality rate approaching 40%. Septic shock is estimated to affect nearly half a million patients in the USA annually, with similar rates observed in Europe. No sepsis-specific treatment exists. There is a broad consensus that sepsis and septic shock are heterogeneous states that might respond differently to therapy based on pathobiological endotypes.^{1–3} Endotoxic septic shock (ESS), characterised by circulating endotoxin, represents a treatable endotype of sepsis.^{4,5} The endotoxin activity assay (EAA) is the only diagnostic test cleared by the US Food and Drug

Administration that can quantify circulating endotoxin.⁵ ESS conveys a greater risk of mortality compared with other forms of septic shock⁶ and constitutes a potential target for personalised sepsis interventions.⁴ In this context, polymyxin B haemoadsorption was developed to remove endotoxin from the circulation. In a post-hoc analysis of the EUPHRATES randomised trial,⁷ patients within the range of treatable endotoxin activity (EAA $\geq 0\cdot60$ and $\leq 0\cdot89$), corresponding to the quantification limits of the assay⁵ and the adsorption limits of polymyxin B,⁸ showed lower mortality compared with controls when treated with polymyxin B in an acute illness severity-adjusted analysis.⁹

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See Online for appendix

Research in context

Evidence before this study

We searched PubMed, MEDLINE, Embase, and ClinicalTrials.gov for randomised trials, observational studies, and meta-analyses published from database inception to Nov 20, 2025, using the terms “endotoxin”, “endotoxic shock”, “polymyxin B”, “Toraymyxin”, “PMX”, “PMX-20R”, “hemoadsorption”, “hemoperfusion”, “septic shock”, and “endotoxin activity assay”, without language restrictions, and reviewed reference lists of relevant trials and reviews. Polymyxin B haemoadsorption has been evaluated in observational studies and randomised trials, including EUPHAS, ABDOMIX, and EUPHRATES. A network meta-analysis suggested a potential mortality benefit (relative risk 0.70 [95% CI 0.58–0.86]), although heterogeneity and overall study quality limited inference. Post-hoc analyses of EUPHRATES identified a biologically plausible subgroup—patients with endotoxin activity assay values of 0.60–0.89—who might benefit from the polymyxin B haemoadsorption therapy, but no previous randomised trial prospectively enriched for this endotype and none of the previous trials prespecified Bayesian incorporation of prior evidence. Across studies, more than 35 000 patients have been screened for trials of endotoxin-targeted therapies, underscoring the rarity of biologically confirmed endotoxic septic shock.

Added value of this study

Tigris is the first randomised trial to prospectively validate the enriched endotype suggested by EUPHRATES—patients with

elevated endotoxin activity and multiorgan failure—and the first study on polymyxin B to use a fully prespecified Bayesian analysis framework to pool evidence from two trials run with the same protocol. The design incorporated an informative prior derived from EUPHRATES while safeguarding robustness through the down-weighting of historical data by 25% and extensive sensitivity analyses. The Bayesian plan included simulation to calculate operating characteristics and showed that this degree of borrowing provided >80% power and acceptable Bayesian error rates. Tigris showed high posterior probabilities of improved survival at both 28 days and 90 days in a prospectively enriched population.

Implications of all the available evidence

Endotoxic septic shock is a high-mortality syndrome (>50% with standard medical treatment) with no alternative endotoxin-targeted therapy. The convergence of results from enriched analyses of EUPHRATES and the prospectively designed Tigris trial supports the hypothesis that polymyxin B haemoadsorption might improve outcomes when both elevated endotoxin activity and multiorgan failure are present. The consistency of findings across timepoints and analytical strategies highlights the importance of biologically informed enrolment and Bayesian methodology for rare, mechanistically defined endotypes of critical illness. Further work is needed to optimise treatment timing, dosing strategies, and validation in broader clinical settings.

To support these observations, we designed and conducted a randomised clinical trial (Tigris) using the same treatment protocol as EUPHRATES,⁷ but with the refined molecular and clinical criteria for enrichment used in the post-hoc analysis.⁹ The alignment of the target populations and protocols, including a standardised test for endotoxin activity, justifies the assumption of an exchangeable treatment effect between Tigris and the EUPHRATES subpopulation and the use of Bayesian analysis to borrow strength from the previous evidence.¹⁰ There is widespread interest in Bayesian analysis to study therapies in complex, critically ill patients, especially in settings where precision medicine is used to select narrow indications. Our primary objective was to examine the safety and efficacy of polymyxin B compared with standard of care (SOC) alone in patients with ESS. We hypothesised that polymyxin B haemoadsorption would reduce mortality in patients with ESS with EAAs between 0.60 and 0.89 units.

Methods

Study design

The Tigris phase 3 trial is a multicentre, open-label, randomised controlled trial, the design of which has been previously described.¹⁰ The study protocol and

statistical analysis plan are provided in the appendix (pp 2–123). The Tigris trial recruited patients from intensive care units (ICUs) at 19 academic and community hospitals across the USA, with sites ranging from small to large. The Tigris trial was registered in 2019 with ClinicalTrials.gov (NCT03901807) and was conducted in accordance with the principles of the Declaration of Helsinki and approved by WCG Institutional Review Board (IRB00000533; study tracking 20191517) on June 13, 2019. This report conforms to the CONSORT 2025 guidelines and published recommendations for the reporting of Bayesian trials (appendix pp 124–126).

Patients

To match the inclusion criteria of the EUPHRATES trial,⁷ adult patients (aged ≥18 years) were enrolled who had: (1) hypotension requiring more than 0.05 µg/kg per min of norepinephrine or equivalent vasopressor support for at least 2 h and up to 30 h before randomisation; (2) received intravenous fluid resuscitation of more than 30 mL/kg within 24 h of eligibility; (3) documented or suspected infection and were receiving intravenous antibiotics; (4) multiorgan dysfunction defined as a Multiple Organ Dysfunction Score¹¹ of more than 9 or a Sequential Organ Failure Assessment score¹² of more

than 11; (5) an EAA result between 0·60 and 0·89 units; and (6) at least one of the following: mechanical ventilation, platelet count less than 150 000 per μL , or urine output less than 0·5 mL/kg per h for at least 6 h despite fluids. Exclusion criteria included an inability to reach or maintain mean arterial blood pressure of 65 mm Hg or higher, the presence of end-stage renal disease requiring chronic dialysis, and clinical evidence of non-septic shock; full exclusion criteria are available in the appendix (p 5). Written informed consent was obtained from all patients or their legally authorised representatives. Data relating to sex, race, and ethnicity were retrieved from medical records.

Randomisation and masking

Eligible patients were randomly assigned in a 2:1 ratio to two groups, polymyxin B plus SOC or SOC alone, via an interactive web-based randomisation system (created by the Contract Research Organization [Beaufort CRO], Norfolk, VA, USA) with a blocked randomisation scheme stratified by site. Research staff at each site were

responsible for enrolment and group assignment. Study personnel, including research staff, the study statistician (GT) analysing the data, and the funder, were masked to allocation. The allocation sequence was maintained by Beaufort CRO independent of the study sites and the sponsor (appendix p 127).

Procedures

Endotoxin activity was measured via EAA (Spectral Medical, Toronto, ON, Canada) on whole blood collected in EDTA (edetic acid) vacutainer tubes according to the manufacturer's specifications. Endotoxin activity is reported as a proportion of the total possible activity (0·00–1·00). A measurement of 0·60–0·89 is defined as high but treatable. An EAA result of 0·90 or higher is considered extreme and unmeasurable due to an asymptotic limit of the assay (appendix pp 129–130).

The investigational device used in this study was the TORAYMYXIN PMX-20R (polymyxin B cartridge; Toray Medical, Tokyo, Japan), an extracorporeal haemo-adsorption cartridge intended for the selective removal

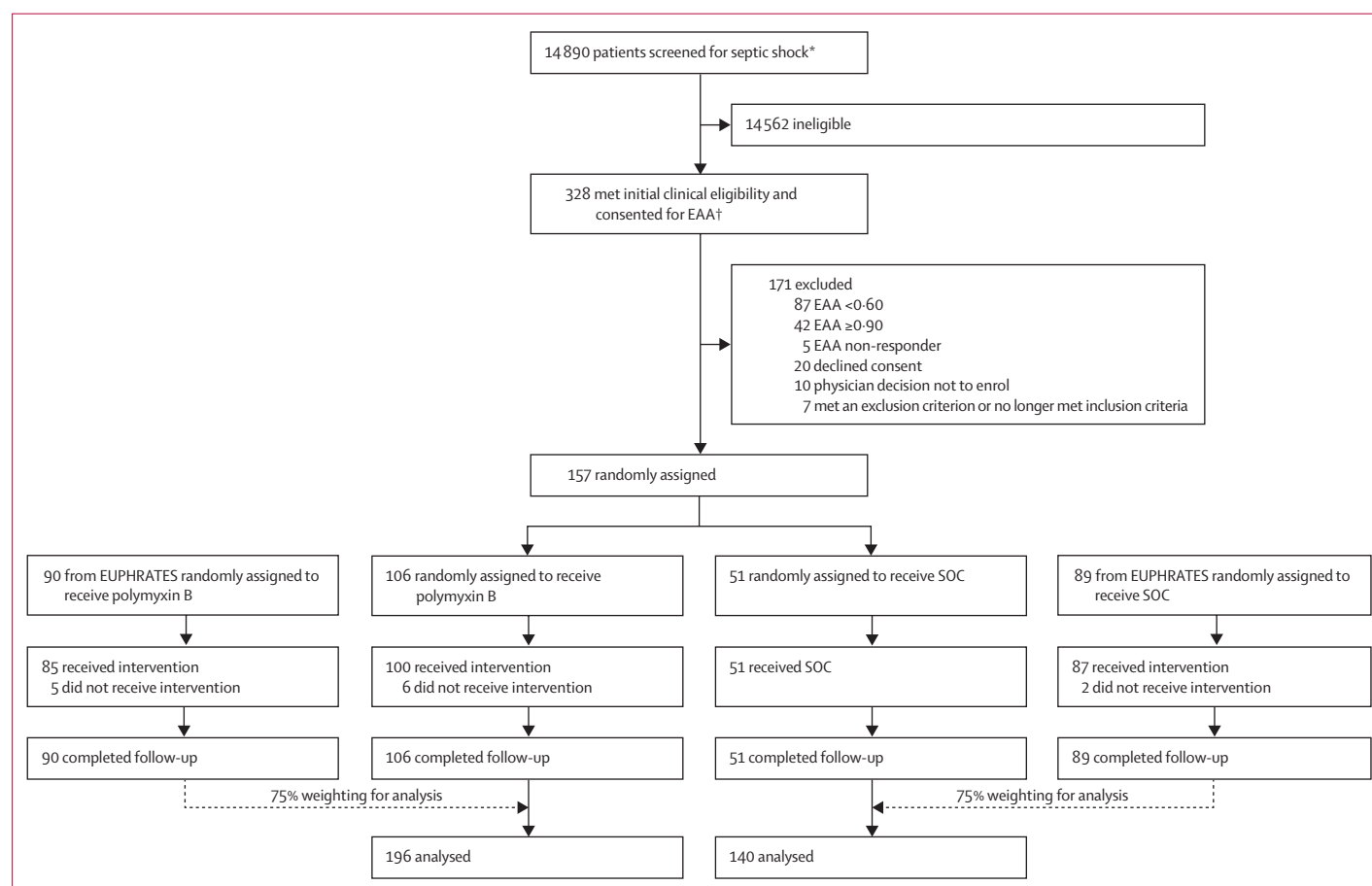


Figure 1: Study flow of patients in the Tigris trial

SOC=standard of care. EAA=endotoxin activity assay. *Pre-screening data were captured between July 1, 2022, and April 10, 2025. There were 11 848 pre-screened patients during that period, leading to 261 patients who met initial clinical eligibility and consented to an EAA. Based on these results, data were extrapolated to estimate an additional 3042 patients who were pre-screened in the period between Sept 30, 2019, and June 30, 2022. †The most common reasons for patient ineligibility were receiving low-dose vasopressors, receiving vasopressor doses outside the protocol-specified duration, having low organ dysfunction scores at baseline, and having clinical evidence for non-septic shock.

of endotoxin directly from circulating blood. The polymyxin B cartridge contains tightly packed polystyrene fibres to which polymyxin B is covalently bonded. Endotoxin has a strong affinity to polymyxin B and binds it through ionic and hydrophobic bonds. These bonds allow the polystyrene fibre–polymyxin B–endotoxin complex to remain within the cartridge during treatment, whereas the purified blood is returned to the patient. Polymyxin B haemoadsorption was delivered with a standard haemodialysis machine or a continuous renal replacement therapy machine at a blood flow rate of 80–120 mL/min, two treatments for 90–120 min each, separated by approximately 22 h. For treatment with polymyxin B, central venous access was obtained with a 12F or 14F double-lumen catheter into the femoral or internal jugular vein. The use of unfractionated heparin as an anticoagulant was recommended during each treatment to prevent thrombus formation within the extracorporeal circuit. The circuit is primed with NaCl 0.9% containing 4 U/mL of heparin, then a bolus of 3000 international units is given systemically to the patient followed by a maintenance of 20 U/kg over the 2 h of haemoadsorption. When renal replacement

therapy was also used it was provided separately from polymyxin B treatment and not in tandem with polymyxin B (ie, intermittent renal replacement therapy was scheduled around polymyxin B and continuous renal replacement therapy was interrupted during the 2-h polymyxin B intervention). All patients continued to receive standard medical care for the management of sepsis as recommended by the Surviving Sepsis Campaign Guidelines.¹³ Follow-up visits or telephone calls were conducted at approximately day 28 and day 90, at which a mortality status was obtained.

Site investigators identified and documented all protocol-defined adverse events and serious adverse events. Changes in medical history or new symptoms, findings, or diagnoses were recorded as adverse events and graded per Common Terminology Criteria for Adverse Events in the electronic case report form for adverse events. Site investigators also assessed each event's relationship to study treatment components and gathered information to determine outcome and status (eg, serious adverse event). Full details are provided in the appendix (p 128).

Outcomes

Because Tigris was designed for regulatory approval in the USA, the primary endpoint was mortality at 28 days after randomisation (data collected by each site). However, the key secondary endpoint was mortality at 90 days and we analysed both timepoints using the same Bayesian statistical models with informative priors. Additional secondary endpoints included mortality at 14 days, survival time from baseline to death within 28 days, changes in mean arterial blood pressure from day 0 to day 3, changes in vasopressor doses, and changes in total duration of use from day 0 to day 3. The primary endpoint was also assessed in patients with a baseline norepinephrine dose $>0.1 \mu\text{g/kg}$ per min. Data on 1-year mortality are also being collected for future analysis. Safety was assessed by comparing rates of treatment-emergent adverse events and all serious adverse events.

Statistical analysis

The primary endpoint analysis of 28-day mortality was conducted on the intention-to-treat (ITT) cohort from Tigris, and the treatment effect was estimated by the odds ratio (OR) comparing polymyxin B to control, adjusting for the baseline Acute Physiology and Chronic Health Evaluation (APACHE-II) in a logistic regression model. The primary analysis used an informative normal prior distribution on the log-OR for polymyxin B based on logistic regression adjusted for APACHE-II at 28 days in the treatable cohort (defined as all patients who were randomly assigned from US EUPHRATES sites with baseline Multiple Organ Dysfunction Score >9 and baseline EAA value ≥ 0.60 and ≤ 0.89). The prior mean was set equal to the estimated log-OR in the treatable

	Tigris		EUPHRATES treatable cohort	
	Intervention (n=106)	Control (n=51)	Intervention (n=90)	Control (n=89)
Age, years	61 (14)	60 (14)	60 (16)	57 (13)
Sex				
Female	47 (44%)	19 (37%)	34 (38%)	35 (39%)
Male	59 (56%)	32 (63%)	56 (62%)	54 (61%)
Race or ethnicity				
White	73 (69%)	36 (71%)	73 (81%)	71 (80%)
Black	17 (16%)	10 (20%)	8 (9%)	8 (9%)
Hispanic	10 (9%)	2 (4%)	6 (7%)	8 (9%)
Asian	2 (2%)	1 (2%)	1 (1%)	1 (1%)
Other	4 (4%)	2 (4%)	2 (2%)	1 (1%)
Arterial blood pressure, mm Hg	74 (12.8)	74.1 (11.6)	71.8 (10.0)	73.3 (10.9)
APACHE-II score	32.8 (8.5)	31.7 (7.3)	32.4 (7.8)	31.1 (8.8)
MODS*	11.6 (1.7)	11.4 (1.7)	12.1 (2.0)	12.0 (1.8)
SOFA score	14.3 (2.0)	14.6 (2.1)	13.4 (2.5)	13.7 (2.4)
Microorganisms				
No growth	41 (39%)	19 (37%)	29 (32%)	28 (31%)
Gram-negative	29 (27%)	15 (29%)	22 (24%)	10 (11%)
Gram-positive	21 (20%)	11 (22%)	20 (22%)	30 (34%)
Mixed	7 (7%)	2 (4%)	12 (13%)	15 (17%)
Other	2 (2%)	0	7 (8%)	6 (7%)
Missing	6 (6%)	4 (8%)	0	0
Bacteraemia	35 (33%)	18 (35%)	30 (33%)	25 (28%)
Cumulative vasopressor index	8.1 (3.3)	7.4 (3.6)	6.6 (3.1)	7.0 (2.7)
Norepinephrine $>0.1 \mu\text{g/kg}$ per min	97 (92%)	47 (92%)	68 (76%)	75 (84%)
Endotoxin activity assay	0.74 (0.07)	0.73 (0.13)	0.73 (0.09)	0.73 (0.08)

Data are n (%) or mean (SD). APACHE=Acute Physiology and Chronic Health Evaluation. MODS=Multiple Organ Dysfunction Score. SOFA=Sequential Organ Failure Assessment. *Only MODS-eligible patients.

Table 1: Baseline characteristics of patients in the Tigris trial and prior distribution

cohort, and the prior variance was set equal to 1 divided by 0.75 times its estimated variance. This 75% weighting discounts the information from EUPHRATES so that the effective sample size represented by the prior is only 75% of the size of the actual sample size.

We ran several Bayesian sensitivity analyses of the primary outcome: a β -binomial model, unadjusted logistic regression, analyses using weights over the range 0% to 100%, and logistic regression models with a commensurate prior that dynamically adjusts the weight based on similarity between the prior and Tigris results (appendix pp 131–133). All analyses were repeated for the key secondary endpoint of 90-day mortality, with prior distributions here based on EUPHRATES 90-day mortality. The analyses were all repeated in the modified ITT cohort (ie, ITT patients excluding those randomly assigned but not treated) and the per-protocol cohort (ie, patients who were randomly assigned and did not have a major protocol deviation), all with appropriate refinements to the prior distributions (appendix pp 128–129). The safety cohort included all patients who were exposed to any amount of study treatment.

The analysis of all remaining secondary outcomes was completed on all randomly assigned patients from Tigris. A linear mixed-effects model was used to compare the mean changes in mean arterial blood pressure and vasopressor doses between arms, with an autocorrelation structure on residuals. Survival time censored at 90 days was summarised with a Kaplan–Meier plot and compared up to 28 days and up to 90 days with Cox proportional hazards models. None of these analyses used prior information, but we also fitted Bayesian Cox models using an informative prior (appendix p 132) on the hazard ratio (HR). Mortality risk at 28 days for the subgroup of patients with baseline norepinephrine dose $>0.1 \mu\text{g/kg}$ per min and mortality risk at 14 days in all patients were summarised by group and absolute risk reductions with treatment were calculated. Adverse events were monitored and reported for all randomly assigned patients for whom treatment was initiated, or who had a proxy time of treatment defined as randomisation plus 2 h (if in the control group).

Using the 75% weighted prior, a baseline risk of 48%, and the distribution of APACHE-II scores from EUPHRATES, we used a simulation study to estimate the chance that, with 150 evaluable patients, the adjusted logistic regression model in Tigris would find at least a 95% posterior probability of benefit (ie, $p(\text{OR} < 1) \geq 95\%$) on 28-day mortality.¹⁴ Here, the posterior probability of benefit is a direct measure, on a 0–100% scale, of the amount of evidence in favour of a reduction in mortality with use of polymyxin B when analyses pool the two sources of information. This Bayesian power was 80% for a true marginal absolute risk reduction of 13.5%. Several considerations entered into the

selection of 75% as the weight defining the prior: it recognised that EUPHRATES provided high-quality, directly relevant evidence; it discounted, by 25%, the precision of the EUPHRATES estimate to reflect uncertainties related to potential small differences in trial populations;^{10,14} it would result in a posterior mean estimate of the log-OR that is approximately a simple mean of the prior value and maximum likelihood value; and the simulation study found that Tigris alone would need to find a clinically meaningful treatment effect (eg, an absolute risk reduction of 6%) in order for the posterior probability of benefit to exceed 95% when Tigris was combined with the prior. Finally, we performed post-hoc detailed analyses to quantify several proposed Bayesian error rates and to test for conflict between the prior and Tigris.¹⁵ Detailed methods can be found in the appendix (pp 134–136).

Role of the funding source

The funder of the study was responsible for the study design, the conduct and management of the study, data interpretation, and the writing of the report, but had no

	Marginal adjusted risk difference* (95% CrI)	Baseline-severity-adjusted OR (95% CrI)	Posterior probability of benefit
28-day mortality, ITT (n=41 intervention, n=23 control)			
Prior	NA	0.59 (0.26 to 1.16)	94.2%
Tigris	6.9% (−9.2 to 23.3)	0.80 (0.38 to 1.49)	79.7%
Posterior	10.3% (−1.7 to 22.3)	0.67 (0.39 to 1.08)	95.3%
90-day mortality, ITT (n=46 intervention, n=31 control)			
Prior	NA	0.63 (0.28 to 1.23)	92.1%
Tigris	18.0% (1.6 to 33.8)	0.50 (0.23 to 0.94)	98.4%
Posterior	15.5% (3.6 to 27.1)	0.54 (0.32 to 0.87)	99.4%
28-day mortality, mITT and safety (n=37 intervention, n=23 control)			
Prior	NA	0.57 (0.25 to 1.14)	94.9%
Tigris	8.5% (−7.9 to 24.8)	0.75 (0.35 to 1.40)	84.3%
Posterior	11.5% (−0.7 to 23.5)	0.64 (0.37 to 1.03)	96.8%
90-day mortality, mITT and safety (n=37 intervention, n=23 control)			
Prior	NA	0.62 (0.27 to 1.20)	92.9%
Tigris	19.2% (2.7 to 35.1)	0.48 (0.22 to 0.90)	98.8%
Posterior	16.6% (4.4 to 28.4)	0.52 (0.30 to 0.84)	99.6%
28-day mortality, per protocol (n=23 intervention, n=18 control)			
Prior	NA	0.51 (0.20 to 1.09)	96.1%
Tigris	10.9% (−6.5 to 28.6)	0.67 (0.28 to 1.35)	88.7%
Posterior	13.7% (0.9 to 26.8)	0.56 (0.30 to 0.96)	98.2%
90-day mortality, per protocol (n=28 intervention, n=25 control)			
Prior	NA	0.58 (0.23 to 1.20)	93.7%
Tigris	20.8% (2.6 to 38.4)	0.45 (0.19 to 0.90)	98.7%
Posterior	18.2% (4.9 to 31.2)	0.49 (0.27 to 0.82)	99.6%

The ITT population includes all patients randomly assigned, with the mITT population excluding those who did not receive treatment. The per-protocol cohort excluded patients who did not receive two treatments of at least 90 min each or the corresponding control intervention. All analyses were performed with a severity-adjusted model with a 75% weighted prior. CrI=credible interval. ITT=intention to treat. mITT=modified intention to treat. NA=not applicable. OR=odds ratio. *Mean risk difference calculated from baseline severity-adjusted model; this calculation involves the distribution of APACHE-II in Tigris, so is NA for the prior.

Table 2: Mortality analysis by endpoint and source

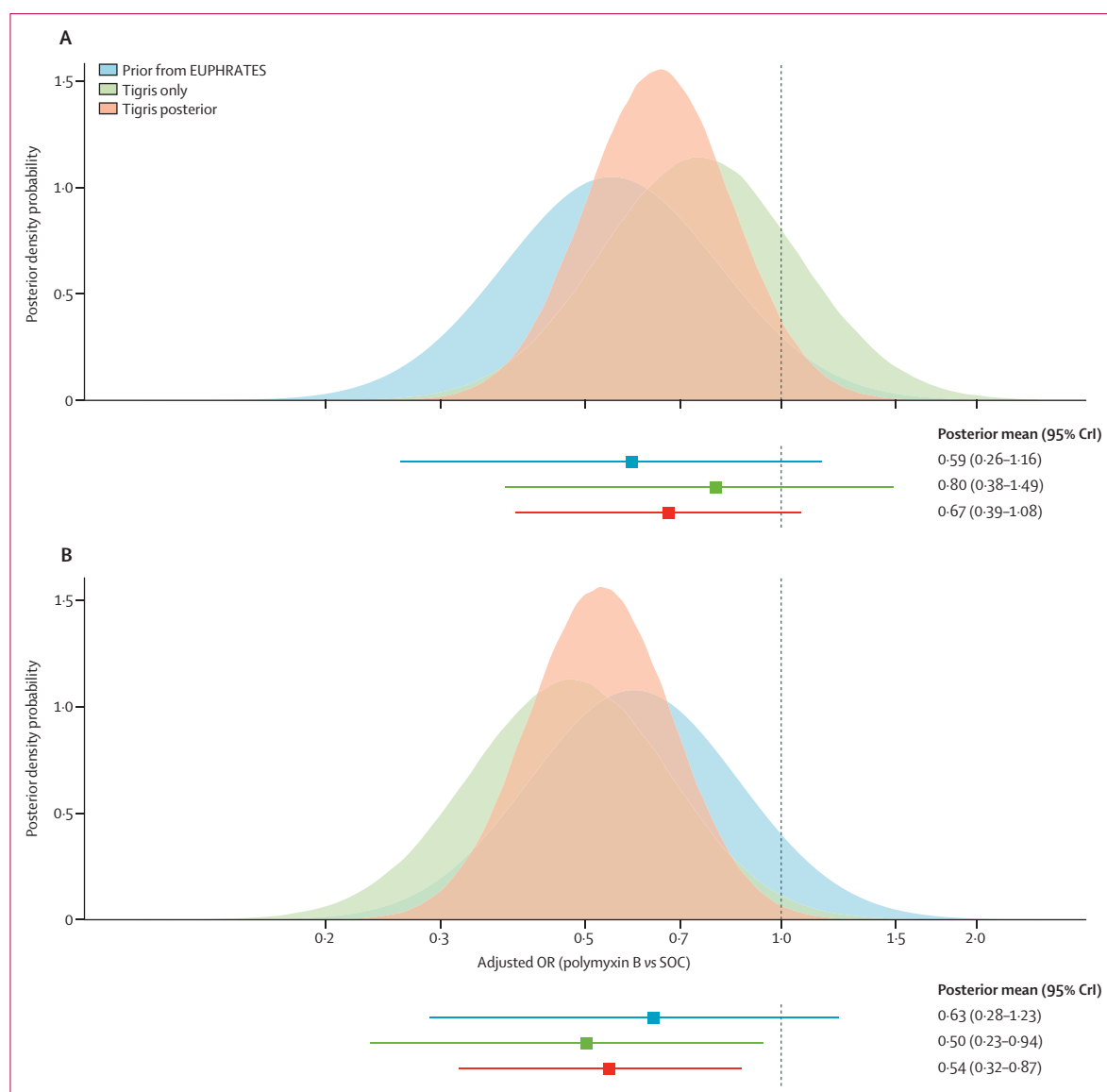


Figure 2: Treatment effects with APACHE-II-adjusted logistic regression models

Tri-plots for the treatment effects of the primary outcome of 28-day mortality (A) and the key secondary outcome of 90-day mortality (B). Each panel has three density plots for the APACHE-II-adjusted OR for polymyxin B treatment: (1) the informative prior distribution based on the 75% weight is shown in blue; (2) the posterior distribution from Tigris only (an analysis using uninformative priors on all parameters) is shown in green; and (3) the posterior distribution from Tigris using the informative prior on the treatment effect is shown in red. APACHE=Acute Physiology and Chronic Health Evaluation. CrI=credible interval. OR=odds ratio. SOC=standard of care.

role in data collection, data analysis, or the decision to submit the paper for publication.

Results

Between Sept 30, 2019, and April 10, 2025, we screened 14890 patients and identified 328 as clinically eligible (figure 1). Of these patients, 194 (59%) had an EAA result between 0.60 and 0.89, and we randomly assigned 106 (68%) to receive the intervention and 51 (32%) to receive the control. Details for modified ITT and per-protocol cohorts are presented in the appendix (p 137). Baseline characteristics are shown in table 1 for all

336 patients in the ITT cohorts: 157 in the Tigris trial and 179 in the treatable cohort from EUPHRATES.¹⁰ Baseline clinical variables were well balanced within cohorts, by treatment group, and across cohorts. Follow-up to 90 days was completed for all patients.

Treatment effects on mortality at 28 days (ie, primary endpoint) and 90 days (ie, key secondary endpoint) are shown in table 2 and illustrated in figure 2. For the primary analysis with a severity-adjusted model with a prespecified 75% weighted prior, there was a 95.3% (adjusted OR 0.67, 95% credible interval [CrI] 0.39–1.08) posterior probability that polymyxin B is beneficial

compared with SOC alone. At 90 days, the same model yielded a posterior probability of benefit of 99·4% (0·54, 0·32–0·87). Similar results were observed for the modified ITT cohort and the per-protocol cohort (table 2). Marginal mean risk reductions for mortality from the adjusted logistic regression model were 10·3% (95% CrI –1·7 to 22·3) at 28 days and 15·5% (3·6 to 27·1) at 90 days (number needed to treat 9·7 at 28 days and 6·5 at 90 days).

We performed prespecified sensitivity analyses on the primary and key secondary endpoints (appendix pp 130, 138–141).¹⁰ The results for a range of alternative weights on the prior are shown in the appendix (pp 138–139). At 28 days, the posterior probability of benefit was above 95% for weights of 72% or greater: 97·1% at 100% weight, 92·4% at 50% weight, 79·6% when using an uninformative prior, and 89·5% when using a commensurate prior. At 90 days, results were essentially invariant to the weight with all posterior probabilities of benefit above 98% even when using an uninformative prior. Results using an unadjusted logistic regression model and using the β -binomial model are shown in the appendix (pp 140–141). These unadjusted models gave lower posterior probabilities of benefit than the fully adjusted model (91·6–92·3% for 28 days and 98·2–98·7% for 90 days) but are qualitatively consistent. Results for 28 days and 90 days with and without adjustment for stratification by site, including a random intercept, showed little difference in any of the estimates, 95% CrIs, or posterior probabilities of benefit (appendix p 141).

We found no evidence of conflict between the prior and the data from Tigris at 28 days or 90 days (appendix pp 144–145); the observed OR in Tigris is not unusually high (at day 28) or low (at day 90) when compared with the values predicted by the priors. Bayesian error rates were low for the design: for example, yielding an estimated 0·7% probability of making an incorrect conclusion of benefit at 28 days (ie, polymyxin B being harmful [true OR ≥ 1] while being found beneficial at the 95% threshold in Tigris; appendix pp 145–146).

Secondary outcomes limited to the Tigris ITT cohort are reported in table 3. Also shown are patient locations on day 28. Of note, larger absolute differences in mortality were observed in patients receiving norepinephrine doses greater than 0·1 $\mu\text{g/kg}$ per min. A greater proportion of survivors were still in the ICU and receiving life support (especially mechanical ventilation) at 28 days in the SOC control group (six [21%], with five [83%] receiving mechanical ventilation) compared with polymyxin B (nine [14%], with seven [78%] receiving mechanical ventilation). None of these patients in the SOC control group were discharged alive from the hospital by day 90 compared with five (56%) of the patients in the polymyxin B group. Furthermore, among the patients still in the ICU on day 28, only one (17%) of six patients survived to 90 days in the SOC control group whereas six (67%) of nine survived to

	Intervention	Control	Difference (95% CrI or CI)
28-day mortality for patients receiving baseline dose of norepinephrine >0·1 $\mu\text{g/kg}$ per min	37/97 (38%)	23/47 (49%)	ARR 10·8 (–6·3 to 27·8)
14-day mortality	36/106 (34%)	19/51 (37%)	ARR 3·3 (–12·3 to 19·4)
Days 0–3*			
Change in mean arterial blood pressure, mm Hg	6·46 (1·53)	6·19 (2·17)	0·28 (–4·99 to 5·54)
Change in vasopressor dose, $\mu\text{g/kg}$ per min†	–0·16 (0·02)	–0·14 (0·03)	–0·02 (–0·01 to 0·06)
Duration of vasopressor use, h	52·3 (25·7 to 69·5)	39·6 (22·6 to 68·1)	11·1 (–10·3 to 21·2)
Location of patient on day 28‡			
ICU§	9/65 (14%)	6/28 (21%)	8 (–8 to 22)
Hospital (non-ICU)	20/65 (31%)	8/28 (29%)	–2 (–21 to 18)
Nursing facility¶	15/65 (23%)	5/28 (18%)	–5 (–22 to 13)
Home	21/65 (32%)	9/28 (32%)	0 (–20 to 21)
Adverse events in mITT and safety sample			
Treatment-emergent adverse events§	57/100 (57%)	28/51 (55%)	–2 (–19 to 14)
Serious adverse events	30/100 (30%)	11/51 (22%)	–8 (–22 to 6)
Events occurring in >2%			
Cardiac disorders	4/100 (4%)	3/51 (6%)	..
Multiple organ dysfunction	6/100 (6%)	2/51 (4%)	..
Infection	9/100 (9%)	6/51 (12%)	..
Respiratory	3/100 (3%)	1/51 (2%)	..
Vascular	4/100 (4%)	0	..
Possibly, probably, or definitely related to intervention**	2/100 (2%)	NA	..

Data are n/N (%), n, least-square mean (SE), or median (IQR). ARR=absolute risk reduction. CrI=credible interval. ICU=intensive care unit. mITT=modified intention to treat. NA=not applicable.

*Analysis restricted to survivors to day 3. †Expressed as norepinephrine equivalents. ‡Percentages of 65 surviving patients in the polymyxin B group (intervention) and 28 surviving patients in the standard-of-care group (control).

§Of the nine patients in the polymyxin B group still in the ICU, six (67%) were alive at 90 days; for the six in the standard-of-care group, one (17%) was alive at 90 days. ¶Skilled nursing or rehabilitation facility. ||Includes only the safety population. **One instance of bleeding from line insertion site deemed to be definitely related to the catheter and one instance of hypotension deemed to be probably related to the polymyxin B cartridge.

Table 3: Outcomes for the Tigris intention-to-treat cohort

90 days in the polymyxin B group. Survival to day 90 (Tigris only, non-Bayesian analysis) is shown in figure 3. The HR up to 28 days was 0·84 (95% CI 0·50–1·40) and up to 90 days was 0·68 (0·43–1·07). We also present the results of a Bayesian analysis using the Cox model with corresponding probabilities in the appendix (p 143). The posterior HR (to 90 days) with informative prior was 0·68 (95% CI 0·47–0·95); posterior probability of benefit 98·8%.

Treatment-emergent adverse events (until day 4) were similar between study groups (table 3). A serious adverse event occurred in 30 (30%) of 100 patients in the polymyxin B group and in 11 (22%) of 51 in the SOC control group (difference –8 percentage points [95% CI –22 to 6]). Two (2%) patients had a serious adverse event that was classified as possibly, probably, or definitely related to the intervention (ie, to the polymyxin B, the

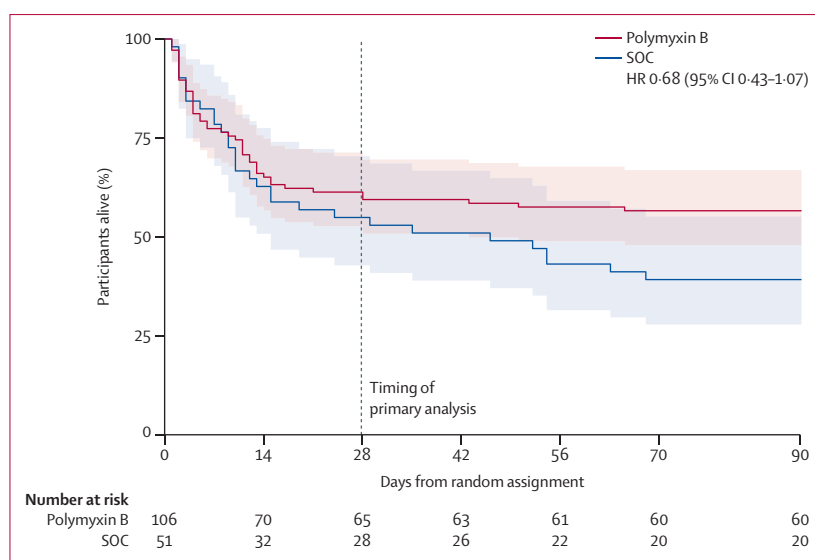


Figure 3: Kaplan-Meier graph of the cumulative survival at 90 days

The solid lines depict the percentage of patients alive in each group and the shaded areas depict the 95% CIs. Results are shown for Tigris only (ie, not including the prior distribution). The HR up to 28 days was 0.84 (95% CI 0.50–1.40). Results for prior distribution are shown in the appendix (p 142). HR=hazard ratio. SOC=standard of care.

catheter, or the anticoagulation); one patient had severe hypotension deemed to be probably related to polymyxin B and one patient had a bleeding event deemed to be definitely related to the dialysis catheter used for polymyxin B. Neither treatment-related event was permanent and both patients were discharged from the hospital (appendix pp 147–148).

Discussion

To our knowledge, Tigris is one of the first trials that has used a precision-medicine approach to sepsis.³ We enrolled patients at particularly high risk, defined by elevated endotoxin activity and multiorgan failure, and showed that polymyxin B haemoadsorption was associated with reduced mortality at 28 days and 90 days, with posterior probabilities exceeding 95% at 28 days and 99% at 90 days. Furthermore, survival over 90 days after random assignment was significantly greater for patients treated with polymyxin B compared with SOC (posterior HR 0.68 [95% CI 0.47–0.95]; posterior probability of benefit 98.8%).

Endotoxin, a lipopolysaccharide of Gram-negative bacteria, is implicated in sepsis pathogenesis,¹⁶ but not all patients with sepsis or septic shock have high circulating concentrations of endotoxin.^{4,6,17} Elevated endotoxin activity thus defines a distinct endotype of septic shock, which might also occur in Gram-positive infections due to gut translocation or other reservoirs.¹⁸

Polymyxin B haemoadsorption became clinically available in Japan in 1994 and in Europe in 1998. Evidence of clinical effectiveness emerged from pooled small clinical trials, observational studies,¹⁹ and analyses of large clinical databases.²⁰ Real-world clinical data suggest

a potential absolute risk reduction of 3–7% in hospital mortality.²⁰ However, there is little definitive evidence from larger randomised trials. The EUPHAS²¹ trial found a significant reduction in mortality as a secondary endpoint. In the ABDOMIX²² trial in France, plus a post-hoc analysis,²³ polymyxin B did not lower mortality, but this trial enrolled a population at much lower risk than that of the EUPHAS trial, with SOC mortality below 20% and low endotoxin concentrations.

The largest trial to date, EUPHRATES, was restricted to patients with septic shock who were found to have an EAA result of 0.60 or higher. Although EUPHRATES did not show a significant improvement in mortality, subsequent work showed that the EAA could not accurately quantify circulating endotoxin when EAA values are 0.90 or higher, and values in this range likely exceed the removal capacity of the polymyxin B cartridge.⁸ A post-hoc analysis of the EUPHRATES trial revealed that in patients with endotoxin activity within the treatable range (ie, between 0.60 and 0.89) who were treated with polymyxin B, mortality at 28 days was lower than in the control group,⁹ motivating the enrichment strategy used for Tigris.

Although mortality at day 28 in Tigris met the predefined criteria for benefit with polymyxin B, $p(\text{OR} < 1)$ was only above 95% when the weight on the prior was 72% or greater, and after adjusting for differences in baseline severity of illness with APACHE-II. By contrast, results at day 90 were robust to model specifications. Note, the posterior probability of benefit is inherently a one-sided comparison, so the equal-tailed two-sided 95% CrI can include the value 1 when the posterior probability of benefit is between 95% and 97.5%, as it is here for 28-day mortality. In addition, mortality differences widened between day 28 and day 90, a pattern consistent with trajectories after critical illness in which persistent organ dysfunction rather than acute shock physiology drives later deaths.²⁴ Patients assigned to standard care were more likely to remain in the ICU with unresolved organ failure at day 28, and a markedly higher proportion of these patients died before day 90 than patients assigned to polymyxin B plus SOC, supporting the observation that the inability to reach early organ recovery is the principal mechanism for death in patients with sepsis.²⁵ Early improvement might therefore shift patients onto a more favourable recovery trajectory, whereas persistent organ dysfunction predisposes to cumulative complications (eg, muscle wasting, nosocomial infection, and cardiovascular complications), which contribute to late mortality. Mortality between 60 days and 90 days has increasingly been adopted as a primary endpoint in septic shock trials because it captures these delayed and recovery-dependent events more reliably than earlier timepoints.²⁶ Sensitivity analyses supported robustness of the benefit seen at 90 days, with posterior probabilities of benefit exceeding 98% even when weak or non-informative

priors were used. Furthermore, modelling of survival up to 90 days with a Bayesian Cox model also showed a high probability of benefit for polymyxin B. Collectively, these findings suggest that interventions capable of reversing organ failure or preventing its persistence might exert a greater effect on survival at 90 days than at 28 days, which might be artificially sustained by intensive organ support during the initial ICU phase.

Adverse events were not increased with polymyxin B. Only 2% of patients had serious adverse events possibly related to the treatment, a safety profile consistent with other extracorporeal therapies such as ICU dialysis.²⁷ Although intravenous polymyxin can be toxic, the bound form within the cartridge is not released into the bloodstream, consistent with safety reporting over decades of clinical use.²⁸

Our study has several strengths. We enrolled a biologically enriched population with strong potential for treatment benefit, closely replicating previous cohort definitions and study procedures.¹⁰ The use of a prespecified Bayesian framework, peer-reviewed in advance, allowed for the incorporation of prior data in a transparent and rigorous way.²⁹ Importantly, results from Tigris replicated key findings from the previous treatable cohort, including baseline characteristics and estimated treatment effects. Finally, we enrolled a population whose race and ethnicity was similar to that of the US population.

Limitations of our study should be acknowledged. Our results are not generalisable to all patients with sepsis or septic shock, but only to the specific endotype studied. Additional research will be needed to understand the potential benefit for polymyxin B outside of the narrow population enrolled in this study. As with most device trials, the intervention was not blinded, but research teams were distinct from clinical care teams and our primary and key secondary outcomes were objective. The trial was not adaptive and the sample size remains modest, reflecting the narrow target population: nearly 35 000 patients were screened across EUPHRATES and Tigris to enrol just 336. The most common reasons for ineligibility were the receipt of low-dose vasopressors or vasopressor doses outside of the protocol-specified duration, low organ dysfunction scores at baseline, and clinical evidence for non-septic shock. Thus, we acknowledge that most ineligible patients were ineligible because they did not have the clinical phenotype of severe septic shock. Of the 328 patients meeting clinical inclusion criteria and consenting to EAA measurement, 134 (41%) had test results outside of enrolment criteria, with 42 (13%) having an EAA value of 0·90 or higher.

Although the use of an informative prior on the treatment effect increases the classic type I error rate, the value of this error rate has been questioned when the prior evidence favours a beneficial treatment effect. We found low values for the proposed Bayesian error rates in our trial design. We also note that at 90 days, the posterior

probability of benefit was more than 98% even when an uninformative prior was used, an analysis that has no issue of inflated classic type I error. We acknowledge there is subjectivity in the choice of the weight to construct the prior in the primary analysis. To address this subjectivity, we showed that the priors chosen for the primary and key secondary analyses were not in conflict with the Tigris-only result, and we also provide tables that allow readers to view the results for any choice of weight. Finally, as in other precision medicine approaches, we selected a very specific subgroup of patients that had both the therapeutic target (ie, high endotoxin activity for predictive enrichment) and a high risk of death from organ failure (prognostic enrichment). Results from a 2025 study suggest that only about one in four patients with septic shock are likely to benefit from therapies directed against endotoxin.⁶ A strategy of enrichment that uses a highly selected population has been applied successfully in other settings, such as cancer.³⁰

In conclusion, the Tigris trial shows a clinically meaningful signal of reduced mortality with polymyxin B in patients with ESS, alongside an acceptable safety profile. Although statistical support at day 28 varies across modelling assumptions, the mortality signal at day 90 remains robust across all sensitivity analyses. These findings underscore the potential of precision medicine approaches targeting endotoxin-defined sepsis phenotypes.

Contributors

Concept and design: DK, DMF, PMW, GT, and JAK. Acquisition, analysis, or interpretation of data: JAN, ML, MAT, AA-K, CG, RR, DD, AT, J-TC, WSB, LWB, NKM, JD, GWW, KBK, KJG, BAM, JE, SK, RK-S, SJ, EK, DMF, PMW, GT, and JAK. Drafting of the manuscript: JAN, ML, MAT, EK, DMF, GT, and JAK. Critical review of the manuscript for important intellectual content: JAN, ML, MAT, AA-K, CG, RR, DD, AT, J-TC, WSB, LWB, NKM, JD, GWW, KBK, KJG, BAM, JE, SK, SJ, DK, EK, DMF, PMW, GT, and JAK. Statistical analysis: GT. Administrative, technical, or material support: EK. Supervision: EK and JAK. Contributors per investigator sites are presented in the appendix (pp 149–151). ML and GT directly accessed and verified the underlying data reported in the manuscript. All authors had full access to all the data in the study and had final responsibility for the decision to submit for publication.

Declaration of interests

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Data sharing

De-identified individual participant data that support these results have been made available as of Oct 1, 2026. Data will be provided contingent upon the approval of a research proposal and the execution of a data sharing agreement (appendix p 152). Requests for access to data can be submitted via the website <https://www.spectraldx.com>.

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